

Villanova University

Department of Mechanical Engineering

ME 4850 SPRING 2026

**FINAL REPORT 4. SYSTEM AND COMPONENT DESIGN OF A POWER
SYSTEM FOR A DEEP SPACE PROBE**

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1. Introduction and Background

This Final Report established a design for a deep-space probe power system to generate electrical power from a radioactive isotope or small nuclear source. The project goal was to minimize the entropy generation throughout the system. This would boost overall system efficiency to convert the highest possible rate of energy into electrical power.

The system operated like a two-temperature engine and utilized a Brayton cycle. As a two-temperature engine, heat rejection was a crucial step in maximizing overall efficiency. This heat rejection was accomplished through the design of a radiator. Other components in the system's thermal management loop included a turbine to generate shaft power, a recuperator, and a two-stage compression component with an intercooler.

The power generation was specified as 50kW and was assumed to be directed right to the gas heater. The heater efficiency was assumed to be 100%, with all the heat being absorbed by the working fluid. The efficiency of a few other components was specified, along with other constraints and values, and is listed below in Table 1.

Table 1. Space Probe Specifications

Parameter	Value
Max Heat Input	50 kW
Turbine Efficiency	65%
Compressor Efficiency	75%
Heat Generator Efficiency	90%
Working Cycle	Brayton Cycle

A diagram of the system is shown below in Figure 1.

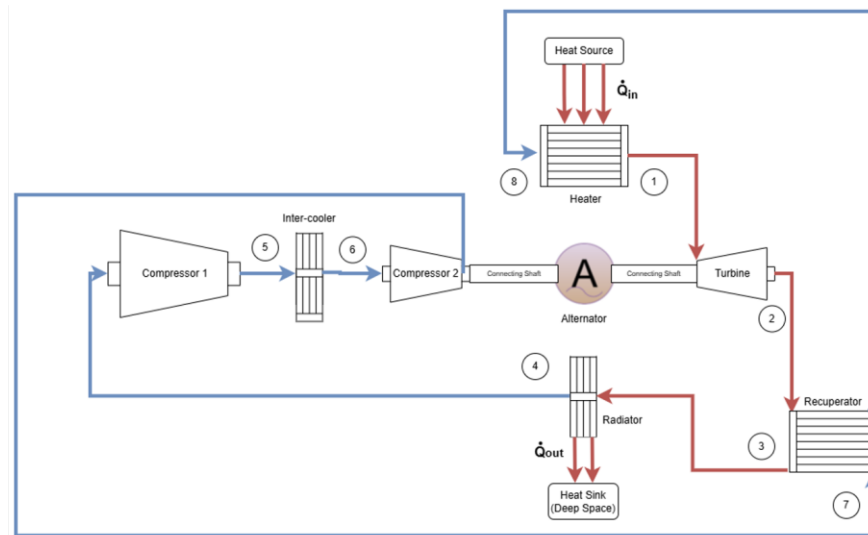


Figure 1. System Diagram

2. Addenda to Interim Report

The first addition to the Interim Report was the justification for the working fluid in the system. The chosen fluid was helium, a change from carbon dioxide used in the Interim Report. Carbon dioxide's critical point was 304.25 K and 7.38 MPa, while helium's critical point was at 5.19 K and 0.22 MPa. The first advantage of helium was that it was much easier to maintain it above the critical point in both temperature and pressure. Another consideration was the fluid's ability to carry heat. Helium's specific heat was around 5.19 kJ/kg-K, while carbon dioxide was only around 0.864 kJ/kg-K. This made helium better at absorbing heat and transferring it through heat exchangers (recuperator, intercooler, and radiator).

Another major consideration was the mass flow rate of the working fluid. The mass flow rate affected values such as the possible pressure drop through the system (even though this was ignored) and the fluid's thermal capacity rate (C), calculated by multiplying the specific heat and the mass flow rate. Higher mass flow rates were more effective at transferring heat, and increased convective heat transfer coefficients, but also increased pressure losses and led to higher entropy generation. This was explored by using the EES code and is expanded on in Section 3 of this report.

One assumption that went unsaid in the Interim Report was the negligence of pressure drops through most components. For example, the assumption was that there were no pressure losses through the gas heater, the radiator, the intercooler, or the recuperator. This assumption eased the design of these elements since they did not have to rely on the pressure drop as a parameter. This also yielded the fluid's enthalpy change due solely to temperature changes, which simplified the calculations for finding the operating points of the helium around the system.

The recuperator and intercooler were both modeled as steady-state control volumes with negligible kinetic and potential energy changes, no shaft work, and negligible pressure drop. For the recuperator, the First Law required the heat released by the turbine exhaust from state 2 to state 3 to equal the heat gained by the compressed stream from state 7 to state 8. For the intercooler, the heat removed from the working fluid between states 5 and 6 was transferred to the secondary loop between states 10 and 9 before being rejected at the radiator.

$$\dot{m}(h_2 - h_3) = \dot{m}(h_8 - h_7) = \dot{Q}_{rec} \quad (1)$$

$$\dot{S}_{gen,rec} = \dot{m}[(s_3 - s_2) + (s_8 - s_7)] \quad (2)$$

$$\dot{m}(h_5 - h_6) = \dot{m}_s(h_9 - h_{10}) = \dot{Q}_{ic} \quad (3)$$

$$\dot{S}_{gen,ic} = \dot{m}(s_6 - s_5) + \dot{m}_s(s_{10} - s_9) \quad (4)$$

These balances were embedded in the EES model so that both component heat duty and entropy generation could be tracked directly during the final optimization study.

3. Optimization Leading to Final Design

Recuperator Design

The recuperator was included because the turbine exhaust still carried a large amount of useful thermal energy after expansion. Rather than sending that energy directly to the radiator, the recuperator transferred a portion of it to the compressed helium before the gas entered the heater. This reduced the external heat that had to be supplied by the radioisotope heat source, lowered the temperature level of the exhaust stream entering the radiator, and improved the overall Brayton-cycle efficiency without adding any moving parts.

A compact counterflow configuration was selected so the hottest exhaust gas always met the hottest compressed stream, and the coldest exhaust gas met the coldest compressed stream, maximizing the average temperature driving force. Two identical cores in parallel were recommended to keep the exchanger package compact enough for the 2 m probe envelope and to reduce flow velocity in each passage. Because both streams were helium and the same mass flow rate passed through each side, the heat-capacity rates were approximately equal, so the counterflow ε -NTU relation for $C_r \approx 1$ was used for sizing. A design effectiveness of 0.80 was selected as a realistic high-performance target that still stayed within the 30–90% parametric-study range required for the project.

$$\dot{Q}_{rec} = \varepsilon_{rec} C_{min} (T_2 - T_7) \quad (5)$$

$$\varepsilon_{rec} = \frac{\dot{Q}_{rec}}{C_{min} (T_2 - T_7)} \quad (6)$$

$$NTU = \frac{UA}{C_{min}} \quad (7)$$

$$A_{rec} = \frac{UA}{U} \quad (8)$$

Using the final cycle mass flow rate of 0.02518 kg/s, a helium specific heat of 5.19 kJ/kg·K, turbine-exit temperature $T_2 = 961.3$ K, and a representative second-stage compressor outlet temperature of about $T_7 = 400$ K, the minimum heat-capacity rate was approximately 0.131 kW/K. The selected effectiveness of 0.80, therefore, gave a recuperator duty of about 58.7 kW and required a total NTU of 4.00. With an assumed overall heat-transfer coefficient of 300 W/m²·K for a compact gas-gas core, the required overall conductance was about 0.523 kW/K, corresponding to a total effective heat-transfer area of approximately 1.74 m². Splitting the design into two identical cores gave about 0.87 m² of area per core.

Table 2. Final Recuperator Design Parameters

Parameter	Value
Working fluid	Helium
Recuperator arrangement	Two identical counterflow cores in parallel
Total mass flow rate	0.02518 kg/s
Hot-side inlet temperature, T_2	961.3 K
Hot-side outlet temperature, T_3	512 K
Cold-side inlet temperature, T_7	400 K

Cold-side outlet temperature, T_8	849 K
Recuperator effectiveness, ε	0.80
Required NTU	4.00
Heat-transfer rate, $\dot{Q}_{rec}$	58.7 kW
Overall U-value	300 W/m ² ·K
Total effective area	1.74 m ²
Area per core	0.87 m ² /core

These results showed that the recuperator performed most of the thermal lifting in the cycle: the compressed helium left the exchanger already close to the heater inlet target, while the turbine exhaust was cooled substantially before reaching the radiator. That trade was why the recuperator was one of the most effective additions made between the interim and final designs.

Radiator Design

A Brayton cycle two-temperature engine required a heat rejection stage, and in this system, that was a radiator dissipating heat from the working fluid to space before the fluid entered the compression stage. The equation that calculated the heat dissipation rate of the radiator was as follows, where ε represented the emissivity of the radiator, σ was the Stephan-Boltzmann constant, A was the radiator surface area, and T_s was the radiator surface temperature.

$$q = \varepsilon * \sigma * A * T_s^4 \quad (9)$$

The designable variables in this equation were the emissivity, based on the radiator material, the area, and the surface temperature, based on the heat transfer system that carried heat from the working fluid to the radiator.

The emissivity was directly correlated with the outer material on the radiator, which was chosen to be a silicon dioxide coating with an emissivity of 0.934.¹ The radiator itself was designed to be aluminum, since it was low-density while maintaining high thermal conductivity.

Heat pipes were chosen to distribute the heat from the working fluid to the radiator surface. Heat pipes used an internal working fluid that was evaporated by the system heat, then condensed to release the heat out of the hot end of the pipe. They made use of wicks to efficiently move the evaporated fluid to the hot end without losing much temperature, which made them a highly efficient and passive method for transporting heat.

The next design point was to determine the necessary surface area of the radiator. This was chosen in conjunction with research into the feasibility of radiator deployment in space and the probe's size constraints (2 meters in diameter). Research showed that many probes of this size deployed 0.5 by 0.5-meter radiators. To maximize the available surface area on the radiators, two radiator panels were included in this system, both double-sided, which brought the total surface area to a total of 1 m².

The EES code returned temperature values for the radiator inlet, which were used to approximate the radiator's surface temperature. Research showed that many heat pipe radiator systems used cold plates with heat pipe cold ends embedded in the base to transfer heat from the working fluid running through the cold plates to the heat pipes.² The surface temperature of the radiator was found by approximating a thermal resistance for the radiator system. Research showed that an approximate thermal resistance value for heat pipes of this scale was around 0.005 K/W. Contact resistances for heat pipe evaporating and condensing to the surrounding surfaces were only around 0.002 K/W. The total thermal resistance was approximated as 0.01 K/W. With this value, the change in temperature from the working fluid to the radiator surface only depended on the total heat travelling through the system, as demonstrated by the Thermal Ohm's Law shown below, where q was the heat rate, and R was the thermal resistance value.

$$\Delta T = q * R \quad (10)$$

The heat pipes for this design were chosen to be evenly distributed along the radiator, to ensure homogenous heating. The total heat through the radiator was determined by plugging values for q into Equation 10 to find a radiator surface temperature, then plugging that value into Equation 9 to find the radiator heat rejection rate until the values aligned. The heat through the system was determined to be approximately 11.0 kW. The temperature change from the working fluid to the radiator surface was 110 K. The heat pipes chosen were 1 cm in diameter and 0.5 m long. 20 pipes laid out in a parallel pattern were spaced 2.5 cm apart to cover the entire radiator surface. For both radiator panels, this totaled 40 pipes. Research showed that heat pipes of this size could carry around 300 W, so the 40 pipes would be enough to transport the 11 kW of energy.³ A visualization of one radiator panel is shown below in Figure 2.

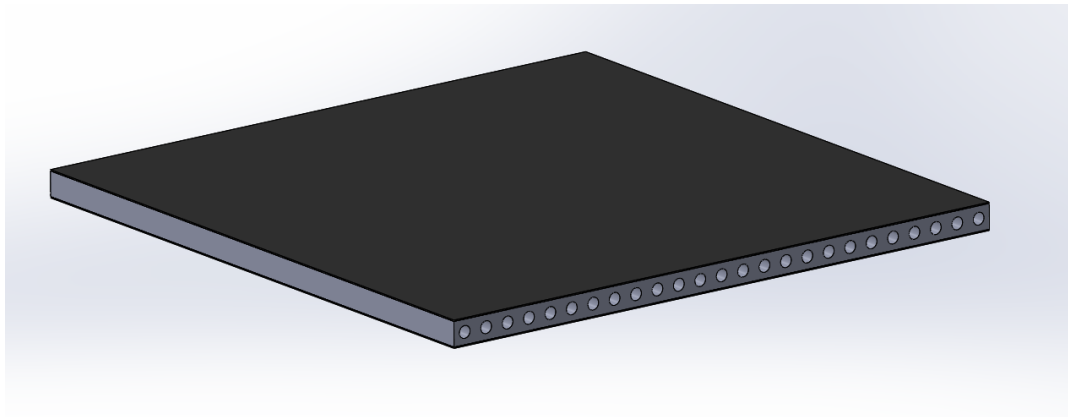


Figure 2. Basic Design of Single Radiator Panel

Intercooler Operation

Another investigation for this report was to determine how intercooling between two compression stages benefited the efficiency of the compression system. The intercooler sought to minimize the temperature of the working fluid leaving the first stage of compression to maximize the pressurization that could occur in the second stage. This worked because a cold working fluid

had a higher density, which meant it could be compressed using less work, making it more efficient. The goal of intercooling was to reduce the temperature of the working fluid entering the second stage of compression to be the same as the temperature it was when it entered the first stage of compression. The intercooler was assumed to be adiabatic, which was important to say that there were no pressure losses through the heat exchanger, only a decrease in temperature to the cooling fluid.

The main design point for the intercooler was at what point in the compression to cool the liquid. Through research, it was found that the optimal intermediate pressure was a result of equalizing the pressure ratios of each compression stage.⁴ The optimal intermediate pressure was calculated using the following equation.

$$P_i = \sqrt{P_1 * P_2} \quad (11)$$

For this design, the intercooler was assumed to have a 95% effectiveness, which research showed was the upper echelon for intercoolers in multi-stage compression systems.⁵ Since this intercooler was for a space probe, it was assumed that it would have been designed to be in the upper echelon of intercoolers considering the likely budget and importance of high-efficiency components.

Engineering Equation Solver Parametric Studies

The existing system EES code was modified to model the intercooler, two separate compressors, and the recuperator. Once this new code included all system components, the parametric study functions were used extensively to generate seven dependency tables. Each table evaluated how altering a single input parameter within a specified range affected three key dependent system values. Each table studied a different independent variable's effect on the system. The independent values studied were mass flow rate, turbine pressure ratio, turbine inlet fluid temperature, recuperator effectiveness, compressor pressure ratio, radiator surface temperature, and the inclusion of intercooling. All seven tables studied the effects on turbine work, compressor work, and overall system efficiency as the dependent variables. The tables were uploaded to an AI software program and prompted to generate seven graphs to represent the tabulated data within the parametric studies it was fed. This allowed more time to be allocated to implementing optimized system inputs in the EES model rather than spending extra time plotting the tabulated data in Excel. The resulting graphs are presented below in Figures 3-9.

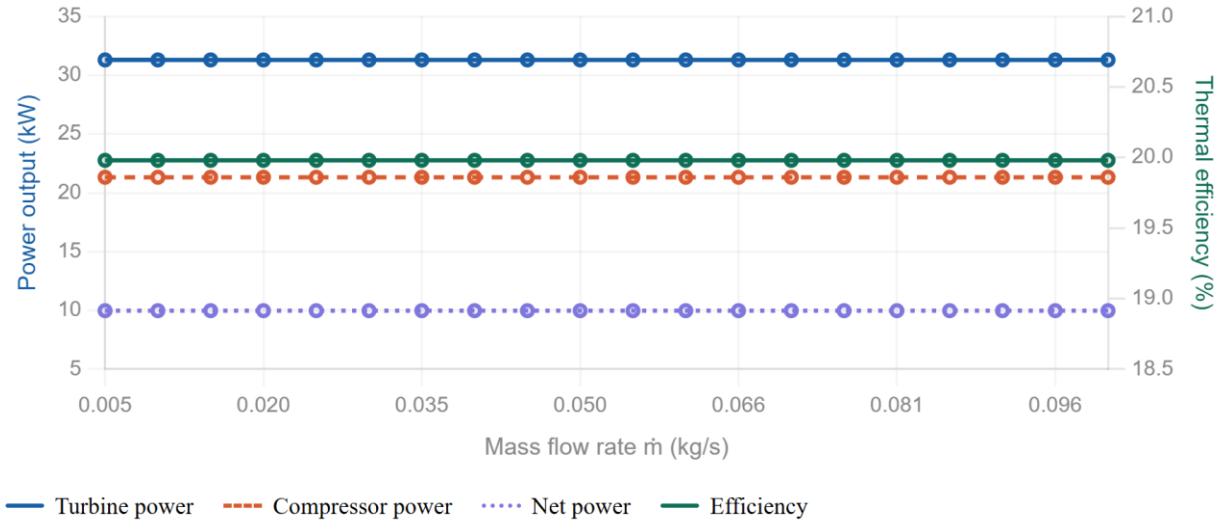


Figure 3. Effect of Mass Flow Rate on Turbine/Compressor Work and System Efficiency

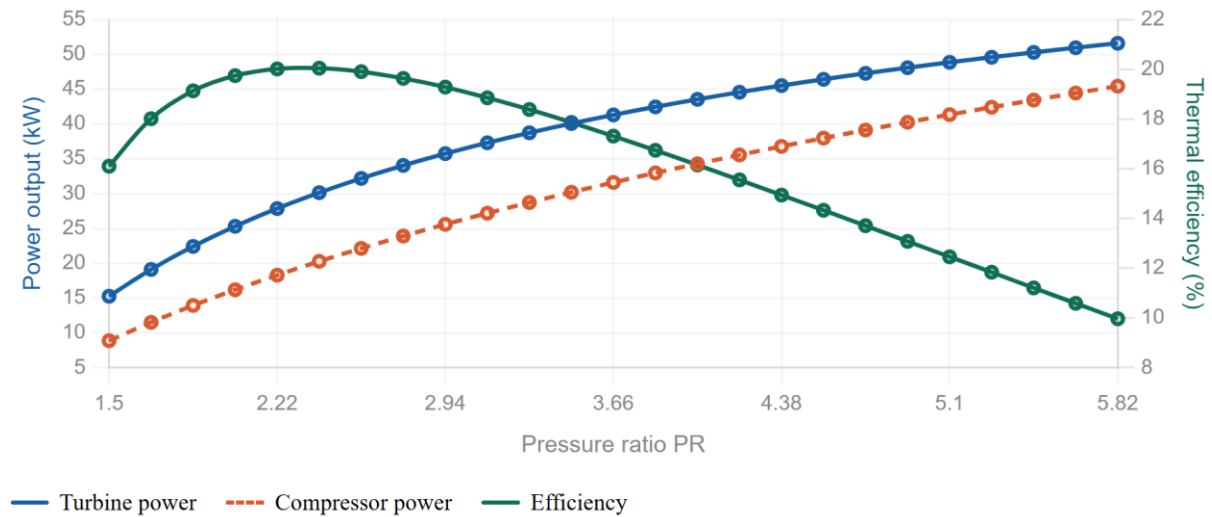


Figure 4. Effect of Turbine Pressure Ratio on Turbine/Compressor Work and System Efficiency

Looking at the efficiency, the optimal pressure ratio was around 2.2, which provided a thermal efficiency of around 19%. This was low compared to research into optimal pressure ratios, but despite troubleshooting the EES code many times, these were the efficiency values and ratios returned. The graph shows that turbine work increases with the pressure ratio, meaning that at higher pressure ratios, the turbine would become more powerful, but the efficiency would decrease.

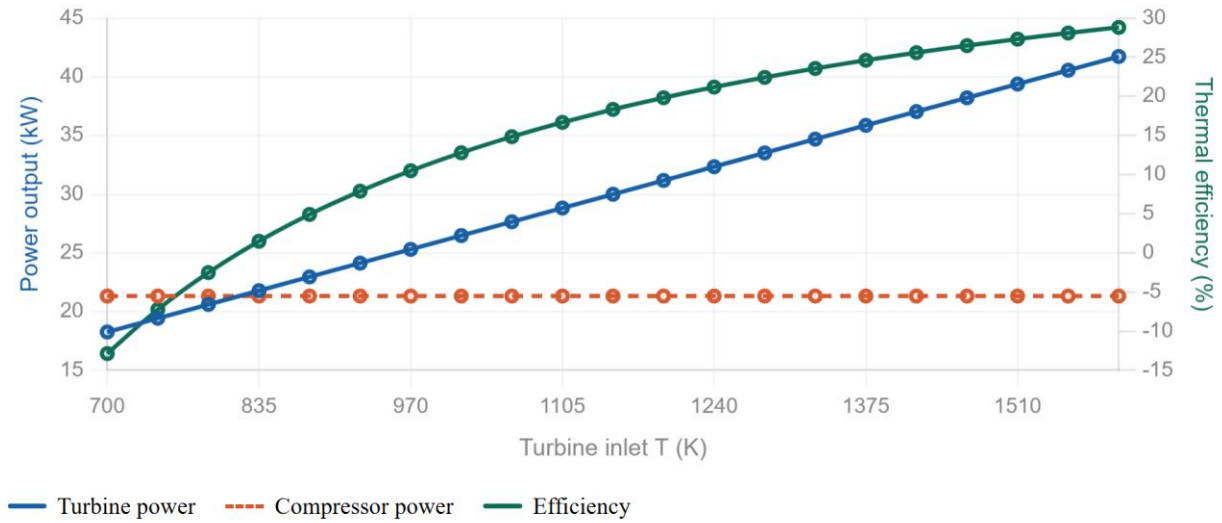


Figure 5. Effect of Turbine Inlet Temperature on Turbine/Compressor Work and System Efficiency

The optimal temperature, as shown in Figure 5, would be as high as possible. As mentioned in the Interim Report, the maximum temperature was tied to the material properties of the gas heater. Titanium was chosen as the material, so the maximum temperature was selected at about 70% of its melting point for safety: 1200 K.

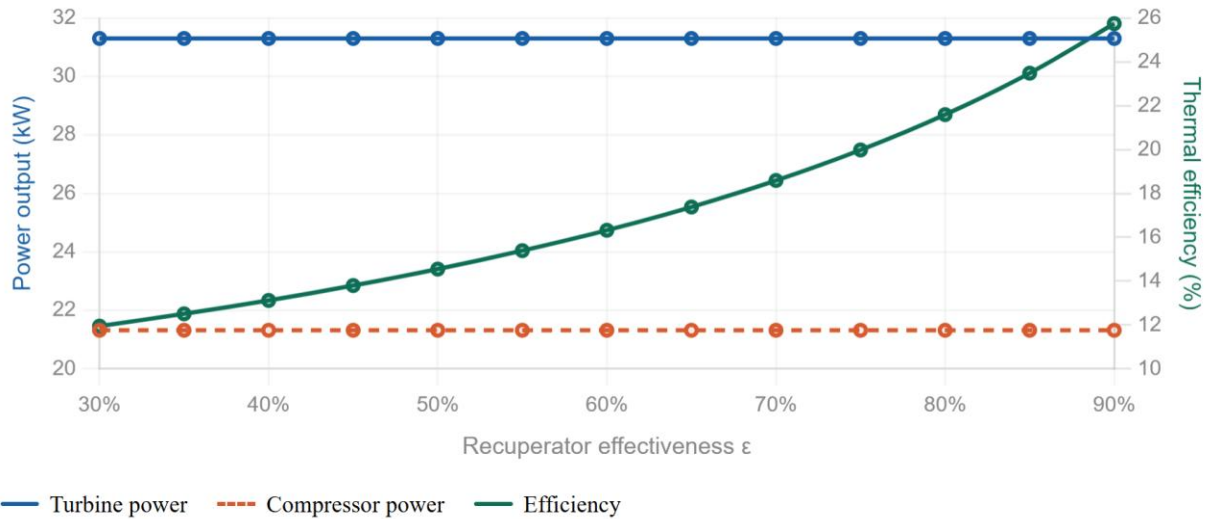


Figure 6. Effect of Recuperator Effectiveness on Turbine/Compressor Work and System Efficiency

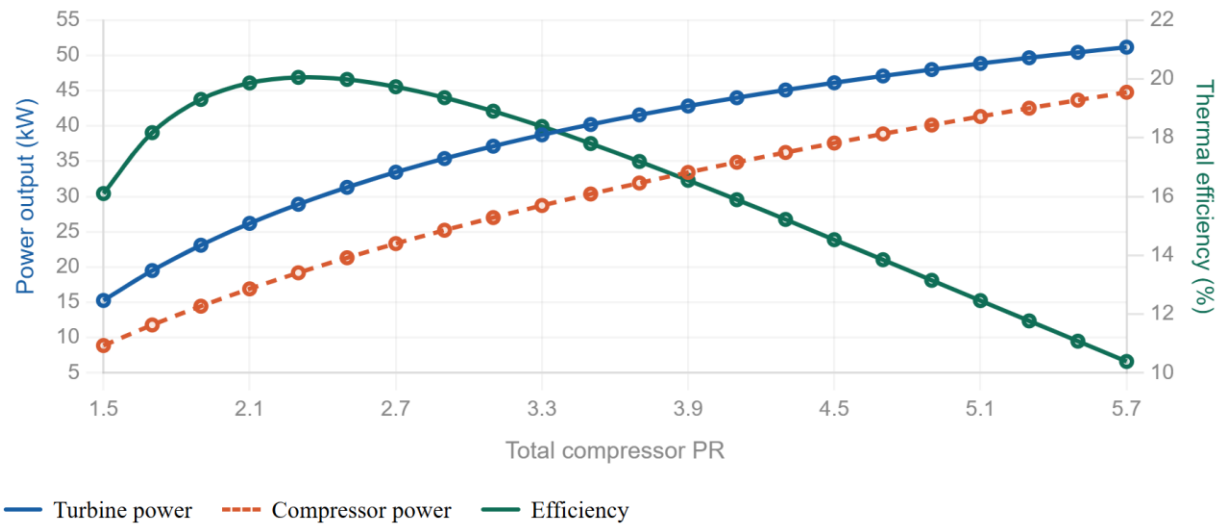


Figure 7. Effect of Compressor Pressure Ratio on Turbine/Compressor Work and System Efficiency

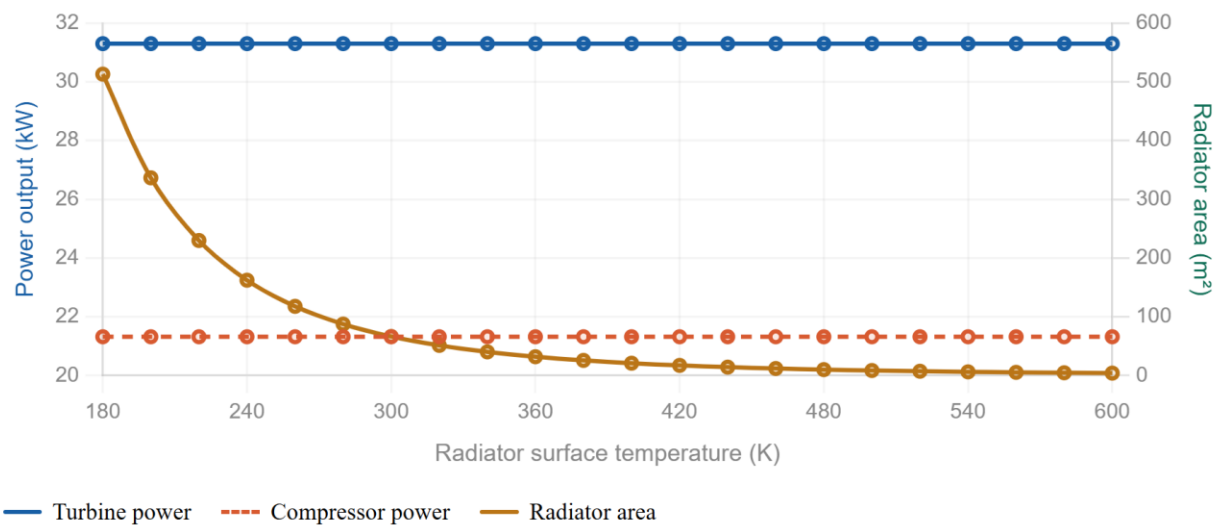


Figure 8. Effect of Radiator Surface Temperature on Turbine/Compressor Work and System Efficiency

Figure 8 matched with the calculated radiator dimensions, considering they were designed for a surface temperature of about 650 K, and a surface area of 1 m².

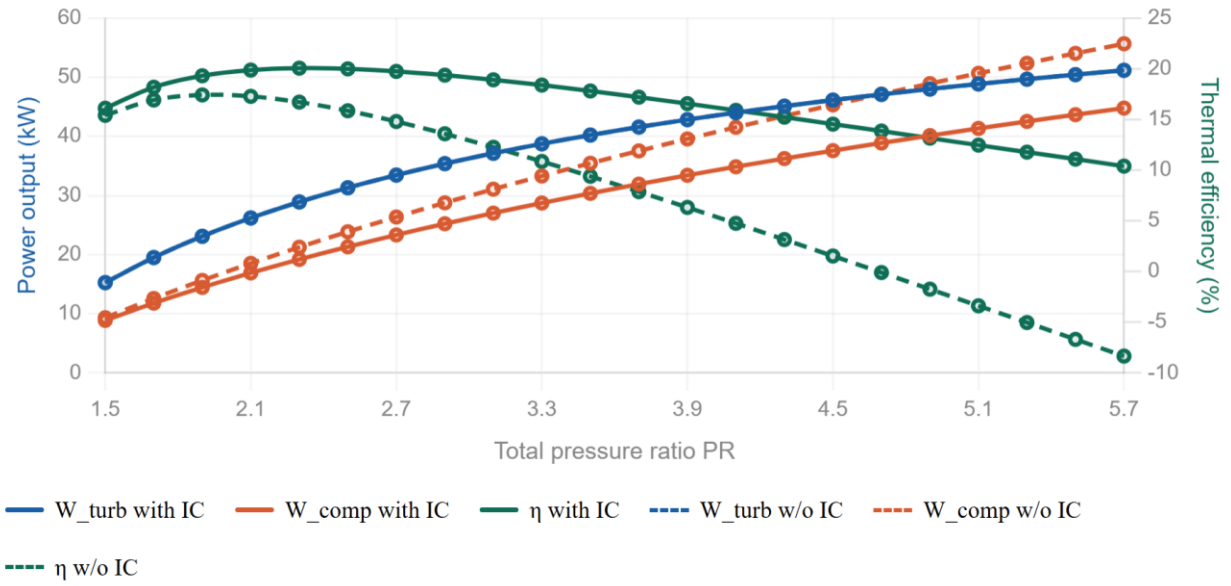


Figure 9. Effect of Inclusion of Intercooler on Turbine/Compressor Work and System Efficiency

Figure 9 demonstrated that the system efficiency was higher with an intercooler, since it increased the working fluid pressurization with less power.

4. Development of Final Design

The Engineering Equation Solver system code [A-1] was used to generate a T-s and P-h diagram for all eight state points, shown in Figures 9 and 10, respectively.

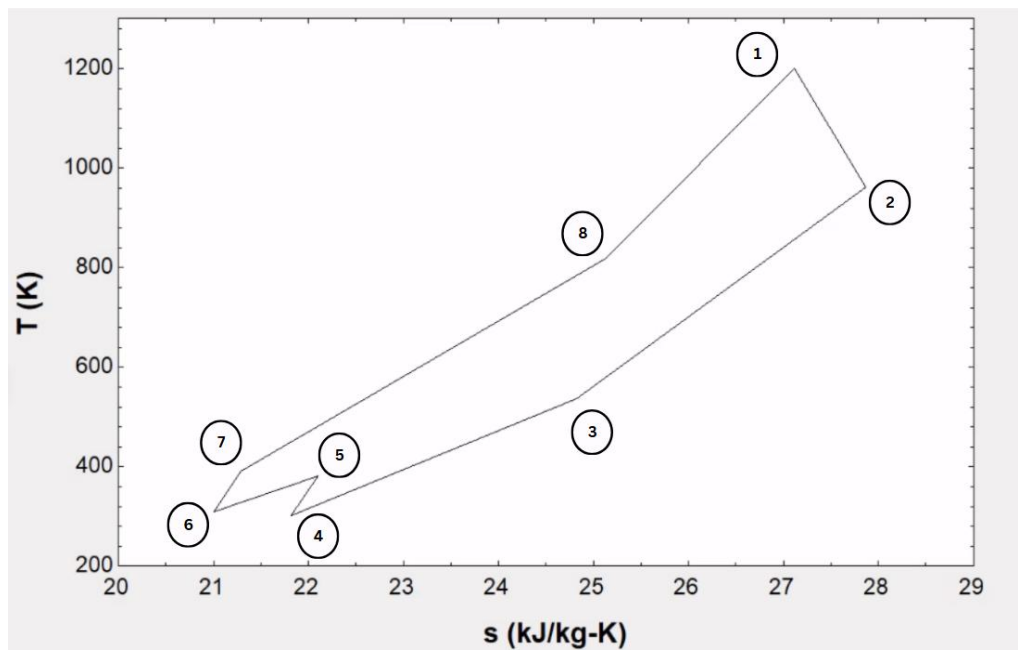


Figure 10. Final System T-s Diagram

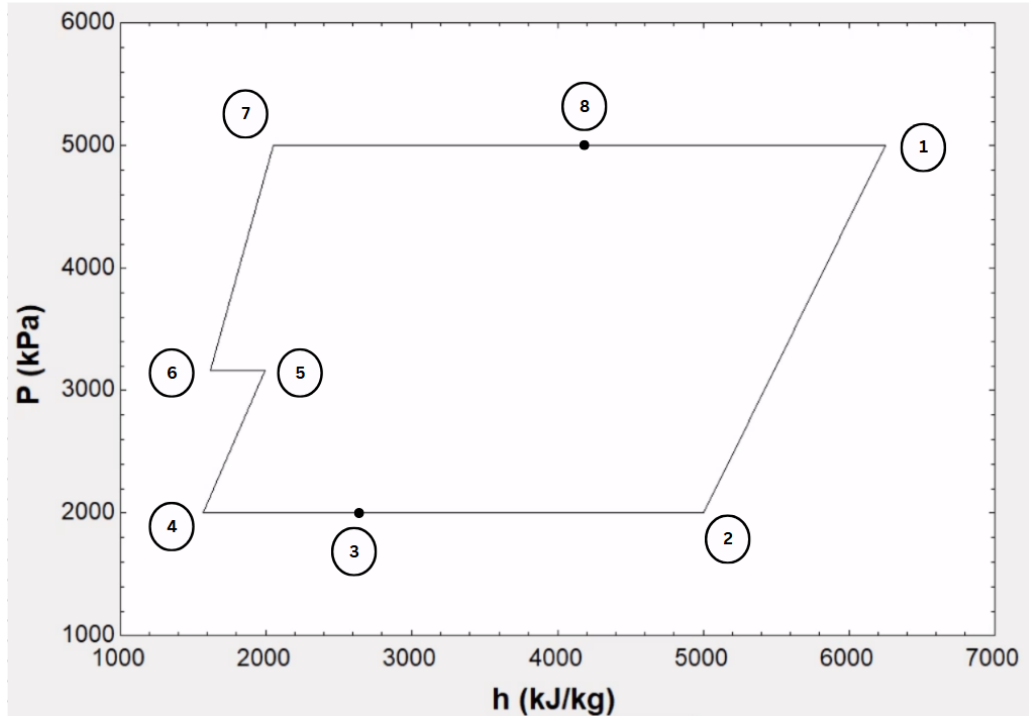


Figure 11. Final System P-h Diagram

Stages 4 to 5 represented the first stage of compression, showing an increase in temperature and pressure due to the compression. From 5 to 6, the intercooler lowered the fluid temperature almost back to the temperature at state 4 without changing the pressure. Then, from 6 to 7, the second stage of compression raised the temperature and pressure again.

In addition to the diagrams, the EES model was used to generate a table of the final system's key states and performance metrics, shown in Table 3.

Table 3. Final System States and Performance Metrics

Parameter	Value
Mass Flow Rate	0.02518 (kg/s)
Heat Input	50 (kW)
Turbine Pressure Ratio	2.5
Turbine Inlet Temperature	1200 (K)
Turbine Inlet Pressure	5000 (kPa)
Turbine Exit Temperature	961.3 (K)
Turbine Exit Pressure	2000 (kPa)
Turbine Work	31.427 (kW)
First Stage Compressor Work	10.628 (kW)
First Stage Compressor Pressure Ratio	1.581
Second Stage Compressor Work	10.978 (kW)

Second Stage Compressor Pressure Ratio	1.581
Generated Electrical Power (Generator)	8.825 (kW)
Overall System Conversion Efficiency	19.61 (%)

The main goal of this project was to limit entropy generation, or the generation of losses, throughout the system. These losses were calculated directly from the Second Law entropy balance equations derived in the Interim Report and expanded on in Section 2 of this final report. The equation to find the rate of irreversibilities ($\dot{I}$) is shown below as a function of the component temperature (T), the energy expelled or consumed by the component ($\dot{Q}$), and the entropy generation rate ($\dot{S}_{gen}$).

$$\dot{I} = \dot{S}_{gen} \pm \left(\frac{\dot{Q}}{T}\right) \quad (12)$$

The irreversibilities in the system accounted for the excess losses in energy that were not converted to electrical power or dissipated by the radiator. These losses were given off as heat in the system, which would normally need to be dealt with by a separate component cooling system. In the case of this space probe, two things were important. First, this design project assumed that the excess heat would not be a problem, and second, the cold shell of the probe would have most likely provided enough cooling to reduce the ambient heat inside the probe.

The entropy generation term in Equation 3 was always a factor of the mass flow rate of the working fluid and the change in the entropy value of the gas through the component, so the only real way to impact the entropy generation was to vary the mass flow rate, which was done in a parametric study to determine the tradeoffs between efficiency and power output of the system.

Equation 3 was used to calculate the irreversibility rate of each component. These values were then tabulated and plotted on a bar chart shown in Figure 12.

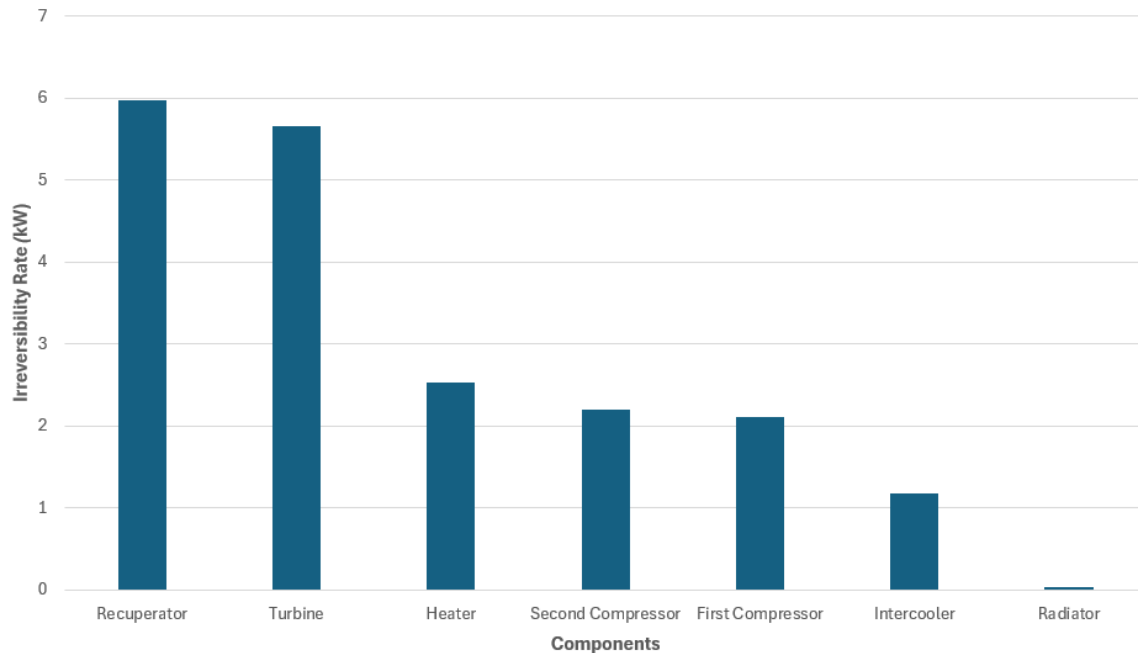


Figure 12. Irreversibility Rate from each System Component

5. Conclusions and Recommendations

This project proved more difficult than expected, due to the use of Engineering Equation Solver as well as the many simplifying assumptions made. While the assumptions did simplify the problem, it made some properties and charts difficult to rationalize. For example, trying to add up the sources of heat loss and energy conversion proved extremely inaccurate since there were many unaccounted-for sources of energy loss, like pressure losses, frictional flow, heat generation, and more.

Another source of difficulty came from trying to design the recuperator and radiator. Many of the values, like inlet and exit temperatures of components, were circular in derivation, meaning they relied on each other. EES was able to solve the system of equations simultaneously, but to design a component a value had to be assumed, then checked, then assumed again to make sure the component fit well in the system. This made it difficult to design when not even the inlet temperature was known until the component was designed and the outlet temperature could be used throughout the rest of the system to find the inlet temperature into the component.

In the end, the code was able to generate a solid Brayton-cycle chart and deduce reasonable state points around the system. Even with the simplifications, the assumptions, and the inaccuracies in the calculations, the general idea of the two-temperature engine held true in the final design, and it was proven to be a functioning design to generate power from the heat of the nuclear source.

References

- (1) Zu, H.; Dai, W.; Li, Y.; Li, K.; Li, J. Analysis of Enhanced Heat Transfer on a Passive Heat Sink with High-Emissivity Coating. *Int. J. Therm. Sci.* **2021**, *166*, 106971. <https://doi.org/10.1016/j.ijthermalsci.2021.106971>.
- (2) *Radiator Demonstration Unit for Space Nuclear Power Systems - Advanced Cooling Technologies*. https://www.1-act.com/resources/learning-center/case-studies/radiator-demonstration-unit-for-space-nuclear-power-systems/?srsltid=AfmBOor3EFVqXk1tpFCZ_kVVyNuYwiwBIF49jE_IllqLi4kbFv5-r6hdi (accessed 2026-04-14).
- (3) *Standard Heat Pipes | MyHeatSinks*. <https://myheatsinks.com/standard-heat-pipe/> (accessed 2026-04-14).
- (4) *Interstage Pressure - an overview | ScienceDirect Topics*. https://www.sciencedirect.com/topics/engineering/interstage-pressure?utm_source=chatgpt.com (accessed 2026-04-14).
- (5) iht-admin. *The Role of Compressor Intercoolers and Aftercoolers in Industrial Applications*. Industrial Heat Transfer. <https://iht-inc.com/the-role-of-compressor-intercoolers-and-aftercoolers-in-industrial-applications> (accessed 2026-04-15).

Appendix

A. Engineering Equation Solver

\$UnitSystem SI K kPa kJ

| "INPUTS"

P1 = 2000 [kPa]
 PR_total = 2.5
 P4 = P1 * PR_total
 PR_stage = **Sqrt**(PR_total)
 P2 = P1 * PR_stage
 P3 = P2
 P5 = P4
 P6 = P4
 P7 = P1
 P8 = P1

T1 = 300 [K]
 T_turbine_inlet = 1200 [K]

eta_c = 0.75
 eta_t = 0.65
 epsilon_recup = 0.75
 epsilon_intercooler = 0.9
 eta_generator = 0.9
 sigma = 5.67E-11 [kW/m^2-K^4]
 T_Space = 3 [K]

"RADIATOR"

h1 = **Enthalpy**(Helium, T=T1, P=P1)
 s1 = **Entropy**(Helium, T=T1, P=P1)

"COMPRESSOR STAGE 1"

s2s = s1
 h2s = **Enthalpy**(Helium, P=P2, s=s2s)
 h2 = h1 + (h2s - h1) / eta_c
 T2 = **Temperature**(Helium, P=P2, h=h2)
 s2 = **Entropy**(Helium, P=P2, h=h2)

"INTERCOOLER"

T3 = T1 + (1 - epsilon_intercooler) * (T2 - T1)
 h3 = **Enthalpy**(Helium, T=T3, P=P3)
 s3 = **Entropy**(Helium, T=T3, P=P3)

Q_intercooler = h2 - h3

"COMPRESSOR STAGE 2"

s4s = s3
 h4s = **Enthalpy**(Helium, P=P4, s=s4s)
 h4 = h3 + (h4s - h3) / eta_c
 T4 = **Temperature**(Helium, P=P4, h=h4)
 s4 = **Entropy**(Helium, P=P4, h=h4)

"RECUPERATOR COLD SIDE"

h5 = h4 + epsilon_recup * (h7 - h4)
 h5 = **Enthalpy**(Helium, T=T5, P=P5)
 s5 = **Entropy**(Helium, T=T5, P=P5)

"HEATER"

T6 = T_turbine_inlet
 h6 = **Enthalpy**(Helium, T=T6, P=P6)
 s6 = **Entropy**(Helium, T=T6, P=P6)

Q_in = h6 - h5

"TURBINE"

s7s = s6
 h7s = **Enthalpy**(Helium, P=P7, s=s7s)
 h7 = h6 - eta_t * (h6 - h7s)
 T7 = **Temperature**(Helium, P=P7, h=h7)
 s7 = **Entropy**(Helium, P=P7, h=h7)

W_turbine = h6 - h7

"RECUPERATOR HOT SIDE"

```
h5 - h4 = h7 - h8  
T8 = Temperature(Helium, P=P8, h=h8)  
s8 = Entropy(Helium, P=P8, h=h8)
```

```
Q_recup = h5 - h4
```

"RADIATOR"

```
Q_radiator = h8 - h1  
T_surface = (T8 + T1) / 2  
A_radiator = (m_dot * Q_radiator) / (sigma * (T_surface^4 - T_space^4))
```

"PERFORMANCE"

```
W_comp1    = h2 - h1  
W_comp2    = h4 - h3  
W_compressor = W_comp1 + W_comp2  
W_net      = W_turbine - W_compressor  
eta_thermal = W_net / Q_in
```

```
BWR = W_compressor / W_turbine
```

"MASS FLOW RATE"

```
Q_source = 50 [kJ/s]  
m_dot    = Q_source / Q_in  
Power_net = m_dot * W_net
```

```
Power_electric = eta_generator * Power_net
```

```
Q_out_total = m_dot * Q_radiator
```

Figure A-1. EES Code